GEANT4 Project
M1 master NE

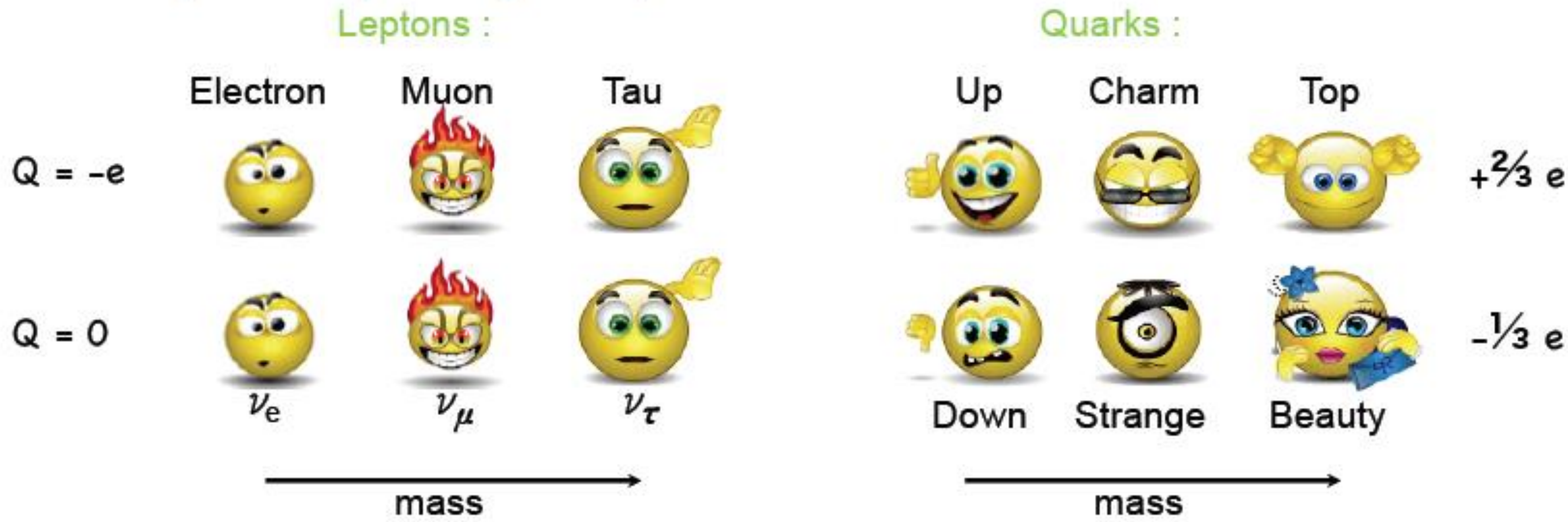
Study of gamma and neutron shielding for the STEREO experiment at the ILL reactor

Items

- Motivation of this work: STEREO experiment @ ILL
- The background problem @ ILL
- Your work step by step
- Expected results

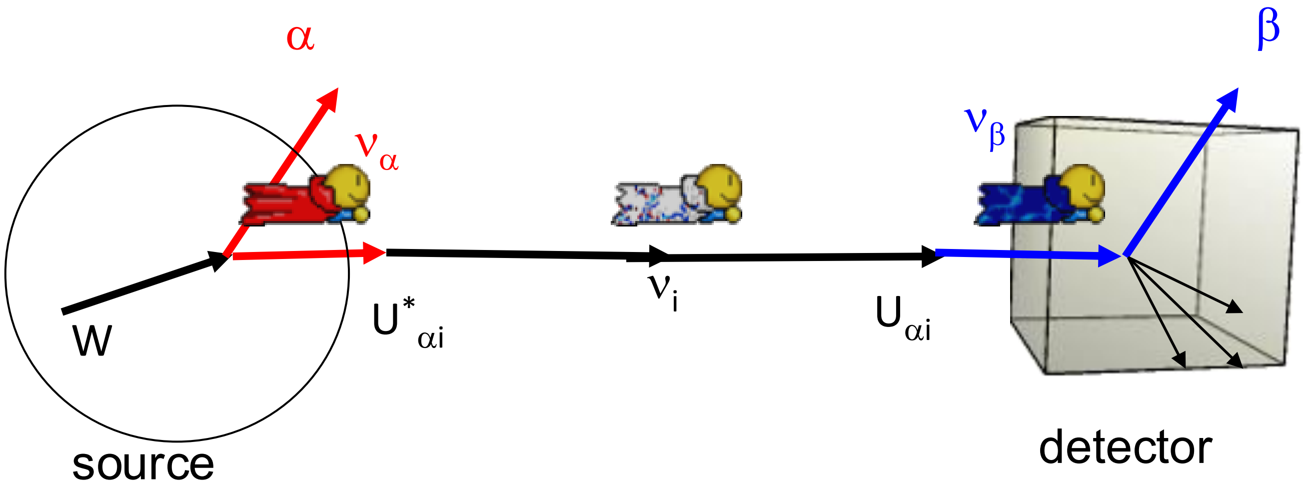
Neutrino in standard model

Fermions (elementary building blocks) :



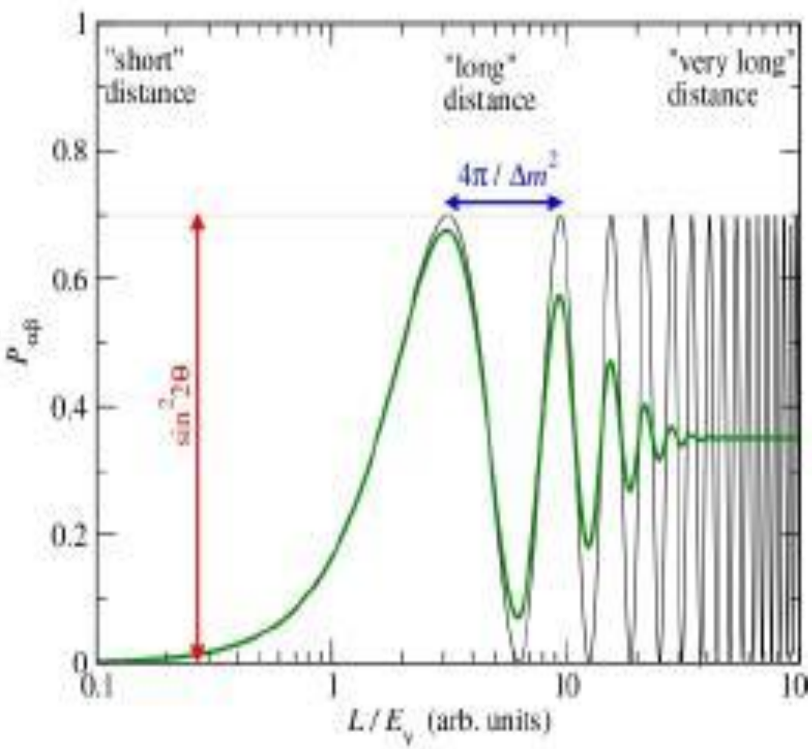
Les interactions :	Intensity	Mediator	Range	Domain
Strong interaction	1	Gluons	1 fm	Nucleus, proton, neutron
Electromagnetism	~ 10^{-2}	Photon	∞	Electricity, chemistry, light
Weak interaction	~ 10^{-6}	Z^0 $W^\pm$	< 1 fm	Radioactivity β, Fusion, sun
Gravity	~ 10^{-38}	Graviton ???	∞	Black holes

Neutrino oscillation



α,β = e, μ, τ $\alpha \neq \beta$
flavor state

i=1,2 or 3 mass state



$$|\nu_\alpha\rangle = \Sigma_i \, U^*_{\alpha i} \, |\nu_i\rangle$$

A neutrino generated as *e* flavor can be detected as μ or τ because it oscillate in a mass state when produced, travel as a mass state and oscillate again in a flavor state that can be different from the source one.

The probability of oscillation depends on the neutrino E and path distance:

$$P_{\nu_\alpha \to \nu_\beta}(L, E) = \sin^2(2\theta)\,\sin^2\left(\frac{\Delta m^2 L}{4E}\right)$$

Oscillation parameters

$s_{ij} = \sin\theta_{ij}$ $c_{ij} = \cos\theta_{ij}$

$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{-i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$

Atmospherics + LBL θ_{13}, CP violation phase Solar + Kamland

- **Oscillation parameters**: 3 mixing angles, 1 phase et 2 mass differences.
- **Present situation (2018)** :

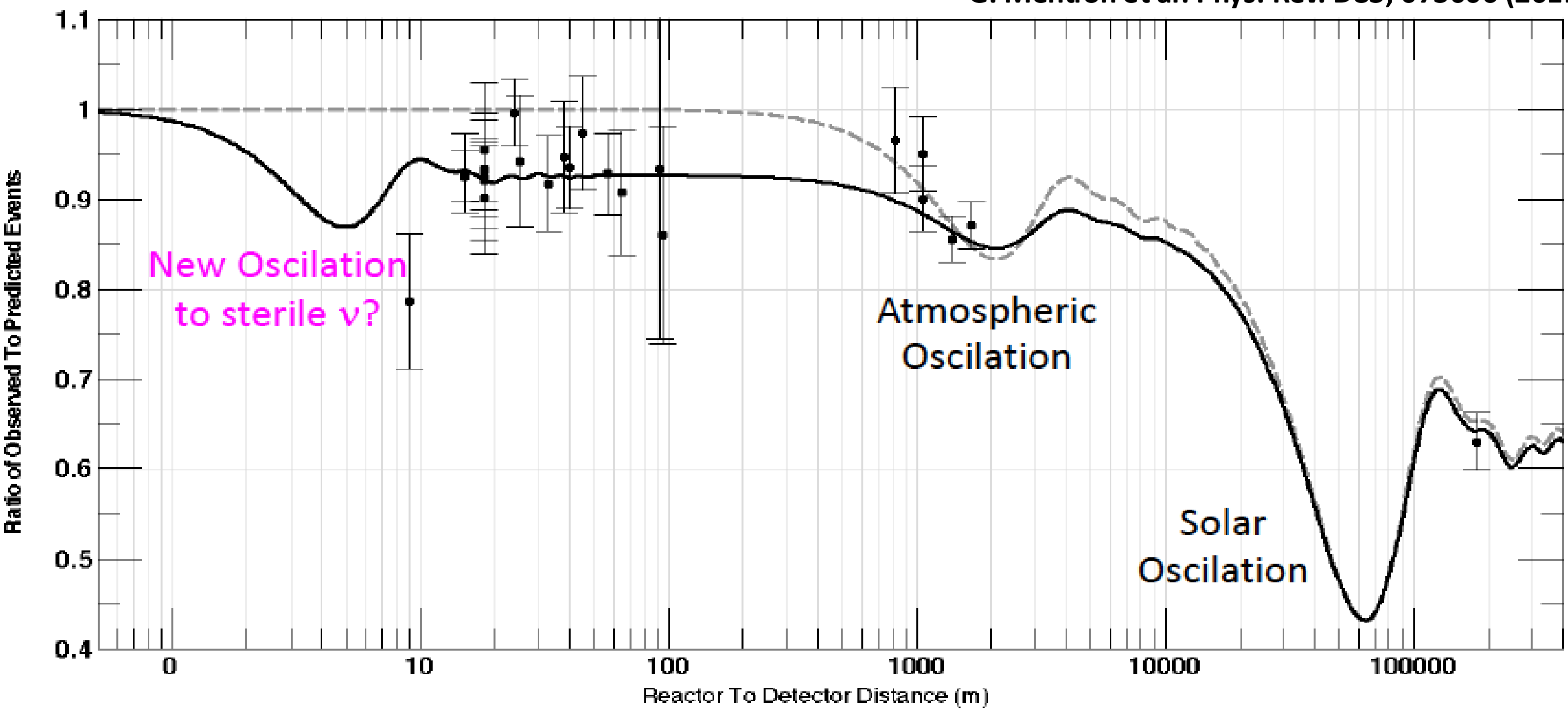
Parameter	best-fit	3σ
Δm^2_{21} [10^{-5} eV ^{2}]	7.37	6.93 – 7.96
$\Delta m^2_{31(23)}$ [10^{-3} eV ^{2}]	2.56 (2.54)	2.45 – 2.69 (2.42 – 2.66)
$\sin^2\theta_{12}$	0.297	0.250 – 0.354
$\sin^2\theta_{23}$, $\Delta m^2_{31(32)} > 0$	0.425	0.381 – 0.615
$\sin^2\theta_{23}$, $\Delta m^2_{32(31)} < 0$	0.589	0.384 – 0.636
$\sin^2\theta_{13}$, $\Delta m^2_{31(32)} > 0$	0.0215	0.0190 – 0.0240
$\sin^2\theta_{13}$, $\Delta m^2_{32(31)} < 0$	0.0216	0.0190 – 0.0242
δ/π	1.38 (1.31)	2σ: (1.0 - 1.9) (2σ: (0.92-1.88))

https://neutrino-history.in2p3.fr/neutrino-oscillation/

IS THERE A STERILE NEUTRINO ?

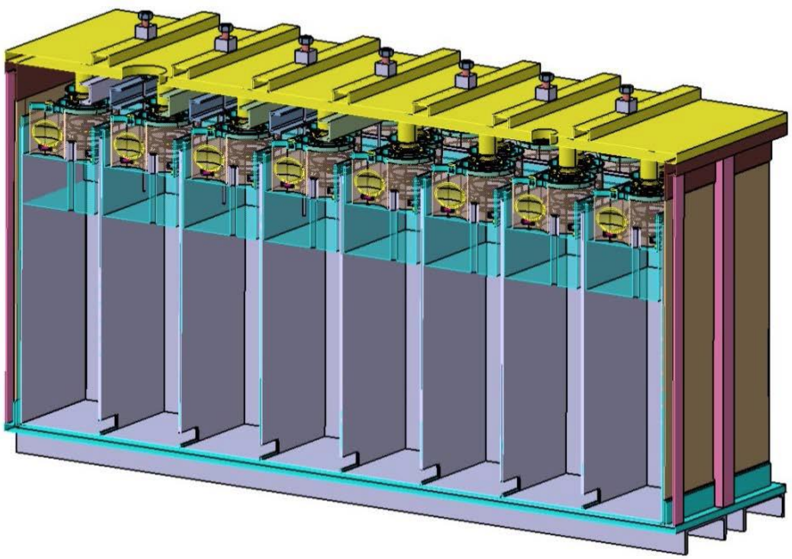
A new analysis give a lack of detected neutrinos at short distance (d>10 m) from reactor, and the oscillation between the 3 known neutrinos cannot explain it → maybe they oscillate in a new kind of neutrino called "sterile" ?

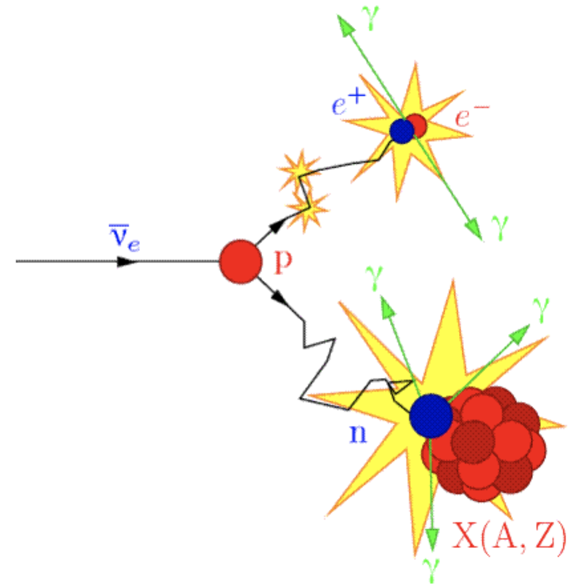
G. Mention et al. Phys. Rev. D83, 073006 (2011)



STEREO

An experiment to study sterile antineutrino at research reactor: a segmented detector filled with Gd doped liquid scintillator





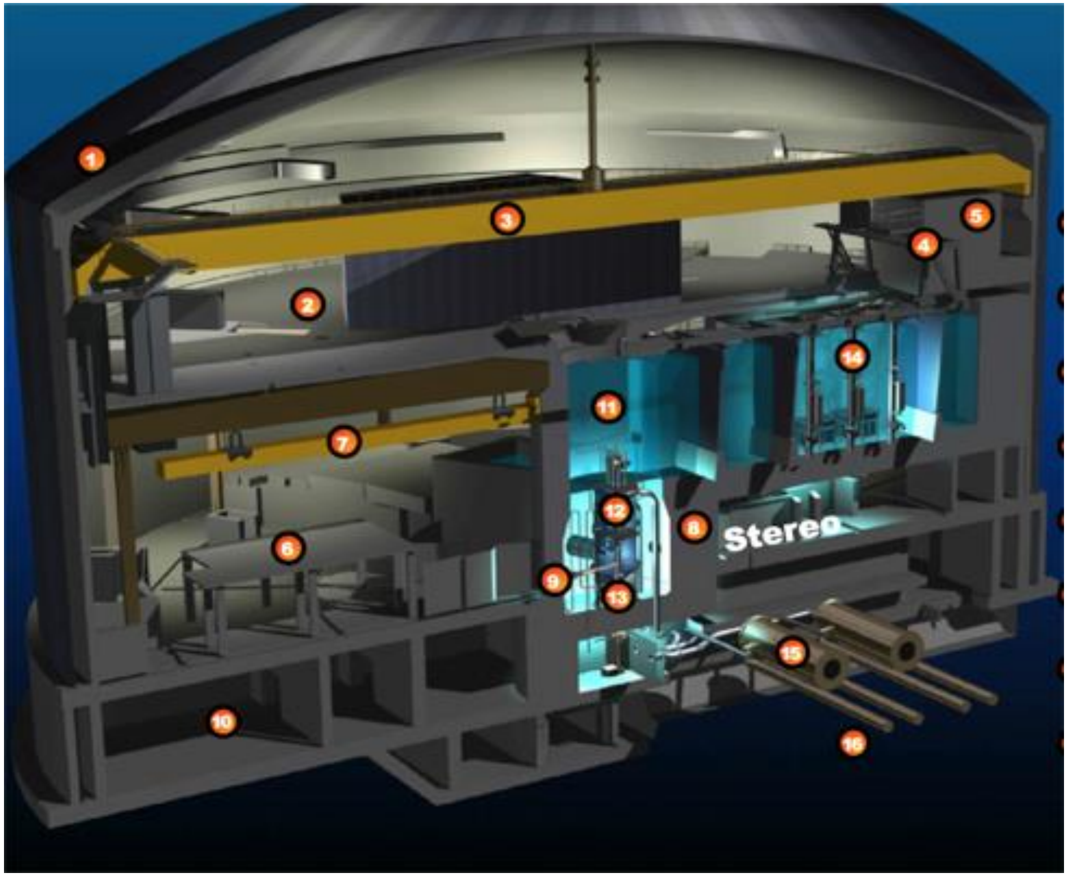
- ν detection via inverse beta decay: $\overline{\nu}_e + p \rightarrow e^+ + n$
- Signature: prompt (e+ annihilation)

AND delayed (n capture on Gd) ~ 15 µs later

$n + Gd \rightarrow Gd + \gamma s$

$e^+ + e^- \rightarrow \gamma\gamma$

ILL research reactor



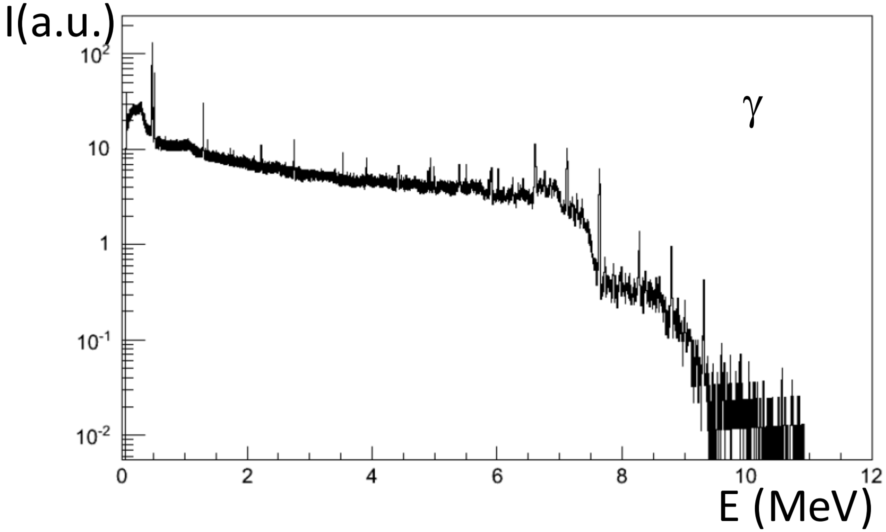
ILL is a research reactor for neutron production and nuclear physics studies (https://www.ill.eu/)

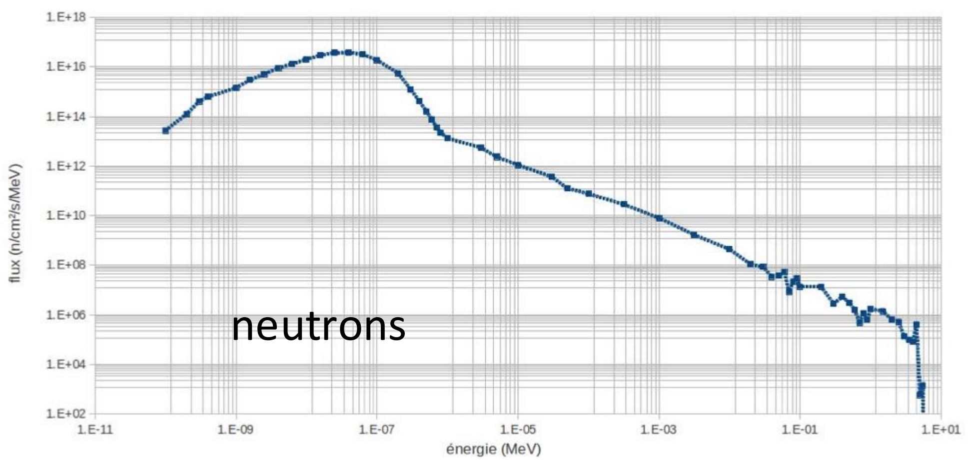
It also produce neutrinos trough beta decay of fission products:

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + e^{-} + \boxed{\bar{\nu}_e} $$

The background problem

ILL reactor emits neutrinos, but also neutrons and gammas



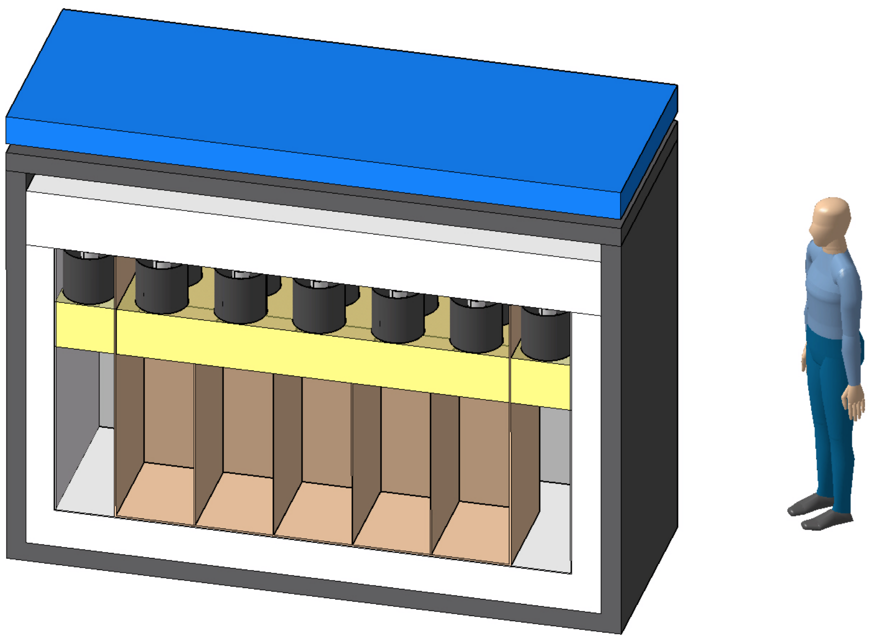


that can mimic neutrino detection:

- Accidental background (gammas and thermal neutrons)
- Correlated background (fast neutrons)

Your work

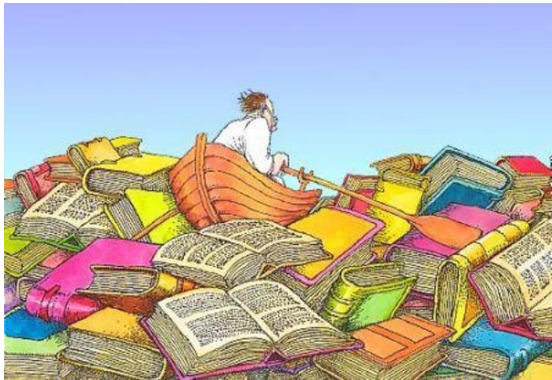
STEREO needs a passive gamma and neutron shielding and you have to check its performances it using geant4 simulation toolkit.



First step: bibliography

We will give you some bibliography indicating:

- Detector geometry and material composition
- Neutron and gammas fluxes, intensities, direction
- How to shield neutrons and gammas
- A first hypothesis of shielding



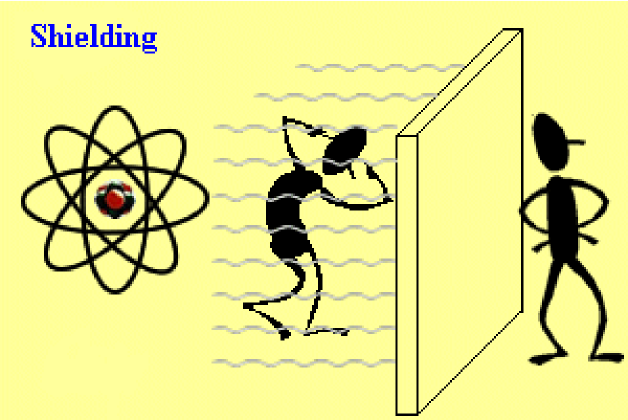
Second step: geant4 simulation

- Take the simulation you already performed at the G4 introduction class
- Integrate a simplified STEREO geometry and the shielding design in the geometry part
- Integrate the information about primary particles you want to simulate
- Verify that your physics list is fine
- Ask for needed outputs
- …….

Third step : verify your shielding

When you are sure your simulation works correctly you can use it to:

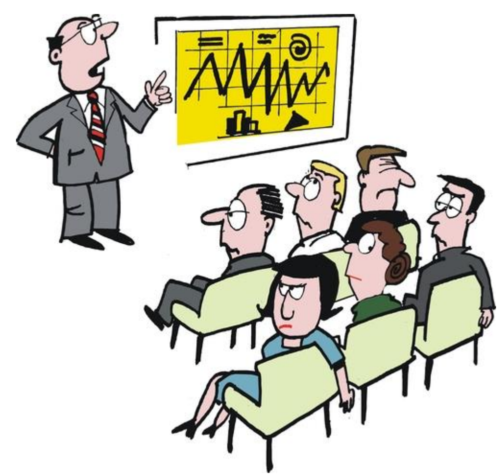
- Verify the efficiency of your shielding
- Compare your background to the expected neutrino rate.
- Calculate your signal/noise ratio
- ……



Evaluation: your code and a report presenting your results

In then report you should presents:

- Your work methodology: how do you proceed, your group organization, eventual problems
- Results of the bibliography: What is Stereo experiment, why STEREO needs a shielding…
- Your simulation: how you implemented the source, the geometry, the approximations you made
- Your results: the energy spectra outputs of your simulation, how you analyzed them
- Your conclusions: what you can learn from your simulation, is the shielding enough for the detection of reactor neutrinos ?



Organization of the work

- You will be in groups of 3 or 4 people, you can chose your colleagues. 3 groups per project.
- About 13 hours in computing class to develop your simulation, extra homework for bibliography and analysis.
- Autonomous work: regular discussion/support by the teachers during the 13 hours but not continuous presence on this time.
- Up to you to organize your work : you can share the work, work together, make regular meetings…